

Shuainan Liu

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EDUCATION

Texas Tech University (TTU)

Ph.D. student (Research Assistant) in Computer Science, Department of Computer Science

Lubbock, USA

Aug. 2023 - Present

- Advisor: Dr. Tianxi Ji
- Research interests: differential privacy, federated learning, diffusion model, cryptography, etc

Sichuan Normal University (SICNU)

M.S in Computer Science and Technology, College of Computer Science

Chengdu, China

Sep. 2018 - jun. 2021

- GPA: 3.53/4 (Top 3 in cohort)
- Advisor: Prof. Chaosheng Feng
- Thesis: Research on File Hierarchy Encryption for CP-ABE
- Core Courses: Advanced Design and Analysis of Algorithms, Number Theory, Stochastic Process, Cloud Computing Technology

Sichuan Normal University (SICNU)

B.S in Information and Computing Science, School of Mathematical Sciences

Chengdu, China

Sep. 2014 - jun. 2018

- Core Courses: Mathematical Analysis, Probability and Mathematical Statistics, Computer Numerical Method, Computer Network Technology

PUBLICATIONS & PATENTS

- **Shuainan Liu**, Tianxi Ji, Zhongshuo Fang, Lu Wei, and Pan Li. 2026. Privacy Loss of Noise Perturbation via Concentration Analysis of A Product Measure. *Proc. ACM Manag. Data* 4, 1 (SIGMOD), Article 66 (February 2026), 27 pages. doi: 10.1145/3786680.
- Shaojie Zhan, **Shuainan Liu**, and Tianxi Ji. Temporal Consistency Gating for Training with Differential Privacy, *Network and Distributed System Security Symposium 2027 (NDSS)*.
- Shixun Li, Maoli Tang, **Shuainan Liu**, Hongjie Zhang, Rong Wang, and Chaosheng Feng. ABE-FL: Efficient and secure federated learning based on CP-ABE with high-security elliptic curves, *Journal of Information Security and Applications*, doi: 10.1016/j.jisa.2026.104636.
- Zhen Guo, Jiangkai Gao, **Shuainan Liu**, Rong Wang, Chaosheng Feng, Keping Yu, Kim Kwang Raymong Choo, and Mohsen Guizani. 2026. *IEEE Transactions on Dependable and Secure Computing*, doi: 10.1109/TDSC.2026.3660271.
- **Shuainan Liu**, Bin Liu, Zhen Guo, Chaosheng Feng, Zhiguang Qin, and Yu Qing. A file hierarchy CP-ABE scheme supporting graded user access. *Journal of Software*, 2023, 34(7):3329-3342.
- Hang Li, Chaosheng Feng, **Shuainan Liu**, Bin Liu, Kaiqiang Zhao. A CP-WABE Scheme Supports Offline/Online Encryption and Verifiable Outsourced Decryption. *Acta Electronica Sinica*. 2020, 48(11): 2146-2153.
- Yunxia Jin, Hekun Yang, Chaosheng Feng, **Shuainan Liu**, Hang Li, Liping Zhou, Gwogen Wan. A KP-ABE Scheme with Outsourced Decryption. *Acta Electronica Sinica*. 2020, 48(3): 561-567.

RESEARCH

Privacy Loss of Noise Perturbation via Concentration Analysis of A Product Measure

Lubbock, USA

Texas Tech University. Supervisor: Prof. tianxi Ji

Aug. 2023 - Nov. 2025

- Designed a novel product noise mechanism for (ϵ, δ) -differential privacy, deriving tighter privacy bounds via geometric measure concentration analysis.
- Integrated the noise framework into output, objective, and gradient perturbations to solve convex and non-convex Empirical Risk Minimizations (ERMs).
- Reduced noise magnitude in high-dimensional tasks, improving utility for convex models (e.g., Logistic Regression, SVM) and privacy guarantees for neural networks.

Security Enhanced File Hierarchy Ciphertext-Policy Attribute Based Encryption

Chengdu, China

Sichuan Normal University. Supervisor: Prof. Chaosheng Feng

Jun. 2019 - Aug. 2020

- Proposed a secure and efficient file hierarchy CP-ABE (SFH-CP-ABE) scheme that eliminating the security vulnerabilities by combining control node and one-way gate, and reconstructing components of key and components of ciphertext.
- Proved completely that the proposed scheme is capable to resist selective chosen plaintext attack (sCPA).

- Theoretical and experimental analysis show that our scheme has high computation and storage efficiency under the premise of improved security while compared to related scheme.

A File Hierarchy CP-ABE Scheme Supporting Graded User Access

Chengdu, China

Sichuan Normal University. Supervisor: Prof. Chaosheng Feng

Sep. 2018 - Jun. 2020

- Construct graded user access tree and redesign the algorithms for encryption and decryption to support graded user access while eliminating the possibility of users conducting unauthorized access.
- The security analysis shows that the proposed scheme is able to resist the selective plaintext attack (sCPA).
- Theoretical and experimental analysis show that the proposed scheme is more efficient in both high computation and storage efficiency compared to the traditional CP-ABE scheme.

WORK EXPERIENCE

Shanghai Pudong Development Bank Co., Ltd., Chengdu Branch

Chengdu, China

Data Analyst, Management Trainee.

Jul. 2021 - Feb. 2023

- Extracted business data from the database and provided analysis services for regulatory reporting, judicial investigation and prosecution, marketing activities, business analysis and other scenarios. Inquired, rectified and reported data that do not satisfy the requirements of the China Banking Regulatory Commission and the head office are carried out.
- Participated in the deployment of several Robotic Process Automation (RPA) projects to meet the needs of business departments for report analysis and information query.
- Participated in the development and testing of several in-line systems, such as the EAST data governance support system and big data platform system.

AWARDS & HONORS

- Expense Support by The Institute for Humane Studies (IHS) (grant no.: IHS020242) (2026).
- Distinguished Graduate Student Assistantships (2023 - 2027).
- Excellent Master's Thesis Award (2020, School Level).
- Third-Class Graduate Student Academic Scholarship (2020, National Level).
- Second-class Graduate Student Academic Scholarship (2018, 2019, National Level).
- Academic Progress Scholarship (2016, School level).

SKILLS & ACTIVITIES

Language: English (Proficient), Mandarin Chinese (Native).

Programming: Proficient with Python, JAVA, C, MySQL, OriginLab, Visio, Microsoft Office.